

Turbulence-Resilient MMF-MCF Interface for FSO-to-Fibre SDM Transmission via Adaptive Wavefront Shaping

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Abstract We demonstrate an interface between free-space OAM-multiplexed link and multi-core fibre link using an MMF-MCF coupling structure with wavefront shaping. We achieve SDM transmission of two 32-GBaud-QPSK channels under turbulence ($D/r_0 \sim 14$), with channel crosstalk reduced by 12.6 dB compared to conventional mode demultiplexing approach. ©2026 The Author(s)

Introduction

Space-division multiplexing (SDM), including mode-division multiplexing (MDM), has emerged as an important technology for expanding optical communication capacity by exploiting spatial dimensions beyond wavelength, time, and polarization [1,2]. SDM has been extensively explored in both free-space optical (FSO) systems utilizing structured beams (e.g., orbital angular momentum (OAM) modes [3,4]) and optical fibre communications employing specialty fibres (e.g., multi-core fibre (MCF), multimode fibre (MMF) [5-7]).

To enable seamless interconnection between SDM-based FSO and fibre links for future optical networks, it is desirable to have an interface that allows spatially multiplexed free-space beams to be efficiently received, demultiplexed, and coupled into SDM fibres [8]. However, atmospheric turbulence can significantly distort optical wavefronts and induces modal coupling [9,10], which severely increases inter-channel crosstalk (XT) and reduces coupling efficiency [11,12].

Prior approaches, including adaptive optics (AO) [13,14] and wavefront shaping [10,15] have been employed to mitigate turbulence-induced impairments. However, most existing demonstrations are limited to demultiplexing free-space channels and coupling them into separate single-mode fibres (SMFs) [10]. A laudable goal might be to realize an interface that can demultiplex turbulence-distorted modes and couple them directly into an SDM fibre, thereby connecting SDM FSO and fibre links in a more integrated manner.

In this work, we demonstrate a free-space-to-fibre SDM interface based on an MMF-MCF coupling structure with adaptive wavefront shaping. By employing the separable natural evolution strategy (SNES) algorithm, we optimize the wavefront shaping and achieve mode-to-core demultiplexing of dual-OAM-multiplexed 32-GBaud-QPSK signals. Furthermore, the large aperture of

the MMF ensures stable received power and robust resistance to turbulence with strength up to $D/r_0=14$. Compared to conventional OAM demultiplexing approach without turbulence mitigation, the proposed scheme achieves an average XT reduction of 12.6 dB.

Concept

As shown in Fig. 1, different OAM modes (ℓ_1 and ℓ_2) carrying independent data channels (Ch1 and Ch2) in an MDM FSO system would suffer from turbulence-induced wavefront distortions and inter-modal XT. In order to separate different free-space modes and couple them into different cores of a MCF for SDM fibre transmission, adaptive wavefront shaping and an MMF-MCF coupling structure are employed. The distorted wavefronts of OAM beams are first shaped by a phase-only spatial light modulator (SLM) and then focused into an MMF. By applying the appropriate phase pattern, the light propagation inside the MMF can be controlled, so that different distorted OAM modes would experience different scattering trajectories inside the MMF. As a result, a mode-to-position mapping can be established to

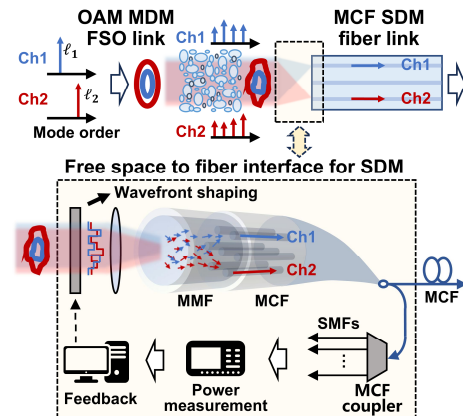


Fig. 1: Schematic diagram of an MMF-MCF coupling structure with adaptive wavefront shaping. The system employs power feedback to optimize wavefront shaping, directing different modes into their respective cores of the MCF.

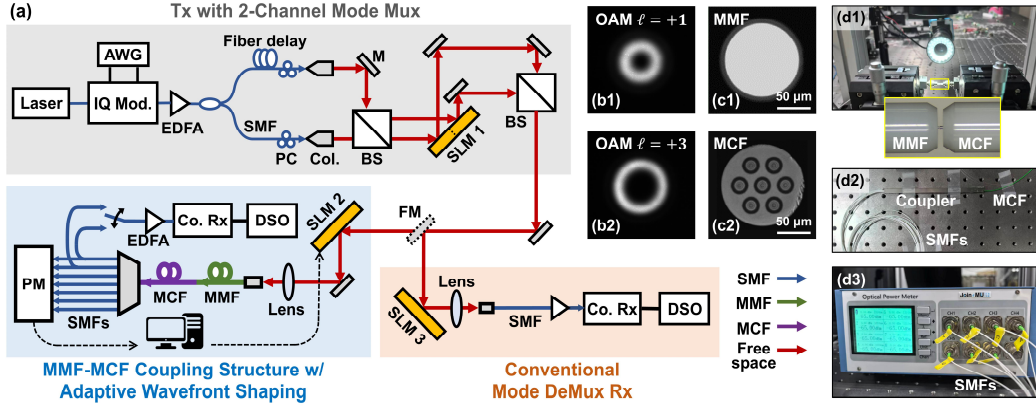


Fig. 2 (a) Experimental setup for the free-space MDM and fiber SDM system. SLM1 for OAM beam generation and turbulence emulating; SLM2 for adaptive optimization based on SNES; SLM3 for conventional Demux; (b1), (b2) intensity profiles of the two OAM modes; (c1), (c2) cross-sectional diagram of the MMF and MCF; (d1) MMF-MCF alignment platform. A pair of six-axis positioning stages is employed to achieve precise alignment; (d2) MCF coupler; (d3) multi-channel PM.

decouple the distinct modes. The modes are then captured by an MMF-MCF coupling structure to achieve the mode-to-core conversion.

To maximize the intensity within the target core while suppressing interference from non-target modes, an adaptive algorithm of SNES [16] is adopted. The search distribution of SNES is parameterized by the mean vector μ and the standard deviation (step size) vector σ . In each generation, N candidates z_k (i. e., a group of phase patterns loaded on SLM2) are sampled and evaluated. The parameters are updated by:

$$\mu \leftarrow \mu + \eta_\mu \sum_{k=1}^N w_k s_k \quad (1)$$

$$\sigma \leftarrow \sigma \cdot \exp\left(\eta_\sigma \sum_{k=1}^N w_k (s_k^2 - 1)\right) \quad (2)$$

where $s_k \sim \mathcal{N}(0, \mathbb{I})$ represents the standard normal samples used to generate candidates $z_k = \mu + \sigma \cdot s_k$, w_k is the utility weights assigned to candidates based on their fitness ranking calculated by the received powers, η_μ and η_σ are learning rates for the mean and step size, respectively.

Experimental setup

Fig. 2 (a) shows the experimental setup. At the transmitter (Tx), a 32-GBaud QPSK signal operating at 1550 nm is generated via an in-phase quadrature modulator (IQ Mod.), amplified by an erbium doped fibre amplifier (EDFA) and split by a 50:50 optical coupler (OC). To ensure decorrelation, one signal path is delayed by a 5 m single-mode fibre (SMF). Both signals are subsequently sent through polarization controllers (PCs) then emitted into free space through collimators (Cols). Using different regions of SLM1, the beams are converted into $\ell=+1$ and $\ell=+3$ OAM beams and combined by a beam splitter (BS) for mode multiplexing. The intensity profiles for the two modes are shown in Fig. 2 (b1) and (b2). Additionally, SLM1 emulates turbulence by adding random

phase to the helical phase [17]. The multiplexed beams propagate in free space for ~ 1 m.

At the receiver (Rx), a flip mirror (FM) is used to steer the light into different mode demultiplexing schemes. For the proposed wavefront shaping and MMF-MCF coupling structure, the received multiplexed modes are first shaped by SLM2 and then focused into a 5 m length of MMF. The MMF output is aligned with a 7-core fibre, as shown in Fig. 2 (d1). The cross-sectional views of the MMF and MCF are shown in Fig. 2 (c1) and (c2), respectively. By monitoring the output powers from the MCF coupler (Fig. 2 (d2)) with a power meter (PM) (Fig. 2 (d3)), the SNES algorithm is employed to optimize the phase pattern on SLM2. For the conventional mode demultiplexing, a conjugate helical phase is loaded on SLM3. This converts the target OAM mode into a Gaussian-like beam. Then, the target signal is coupled into a SMF for coherent detection. Due to only one coherent receiver available, we detect and recover one data channel at a time.

Results and discussion

Fig. 3 shows the received power and XT performance of the two channels, along with the corresponding QPSK constellations and bit error rates (BERs). These results are measured without turbulence effects. Before the optimization of the wavefront shaping (Fig. 3 (a)), the severe XT causes the constellations to be completely scattered, with BERs approaching 0.5, for the multiplexed two channels. By monitoring the output of the MCF, the phase distribution on SLM2 is tuned to maximize the target core intensity. As depicted in Fig. 3 (b), by alternating the optimization of phase patterns among the input modes, the target core powers increase significantly to about -20 dB and eventually stabilize. The optimized phase patterns with 72×72 elements are shown in Fig. 3 (c). Notably, monitoring two selected channels yields comparable performance to

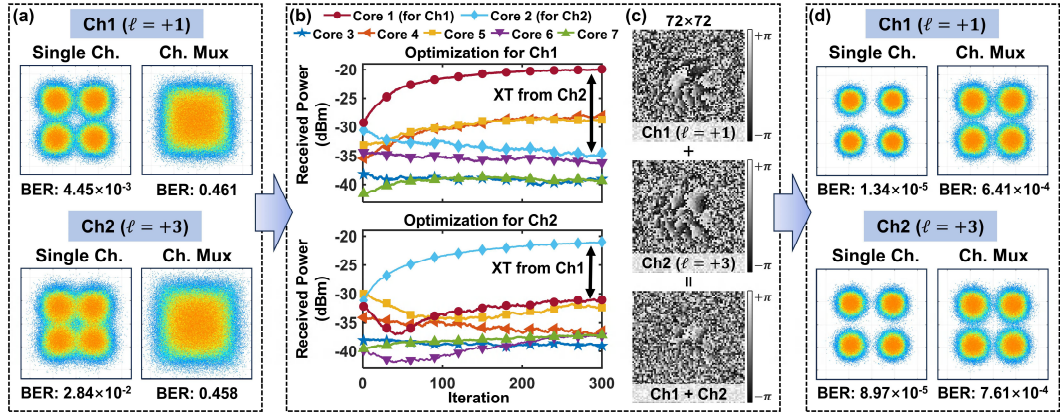


Fig. 3: System performance improvements achieved by the SNES algorithm without turbulence. (a) Baseline XT, constellations, and BERs before optimization. (b) Power convergence curves monitored at the MCF coupler. (c) Phase patterns with 72×72 elements loaded on SLM2. (d) Final XT, constellations, and BERs after WSD-based crosstalk suppression.

monitoring all seven cores, differing only in the convergence time. The final measured XT levels for Ch1 and Ch2 are -14.6 dB and -10.1 dB, respectively. Therefore, after the optimization of the wavefront shaping (Fig. 3 (d)), the BERs for both multiplexed channels drop from approximately 0.5 to 6.41×10^{-4} and 7.61×10^{-4} , respectively.

Fig. 4 illustrates the system performance under varying turbulence strengths ($D/r_0 \sim 7$ and ~ 14). Here, D denotes the beam diameter and the Fried parameter r_0 characterizes the atmospheric coherence length [10,11], with a smaller value of r_0 indicating more severe turbulence. The top row displays the mode intensity profiles under three conditions, showing that the OAM modes lose their ring structure under strong turbulence. The middle and bottom rows compare the XT and BER performances between the conventional Rx and the proposed scheme. Under strong turbulence, the conventional Rx fails to demodulate signals due to severe power fluctuations and XTs. In contrast, the proposed scheme

achieves an average received power improvement of 3.15 dB and XT reduction of 12.6 dB, thereby ensuring the BERs are below the soft-decision forward error correction (SD-FEC) threshold for all the cases. These results show that the proposed scheme maintains robust performance under turbulence.

We note that, with high-speed (over 10's kHz) digital micromirror devices (DMDs) [2] for wavefront shaping and FPGA-based real-time optimization [18], the system is expected to further enhance response speed, enabling real-time tracking and mitigation of fast-varying turbulence. The current coupling setup allows for alignment flexibility and can be further integrated via fusion splicing of the MMF and MCF, which may also further reduce the MMF-MCF insertion loss. Moreover, the system can support various other orthogonal bases, including Laguerre-Gaussian (LG) or Hermite-Gaussian (HG) modes [17,19], and can be potentially scaled to support more spatial modes by utilizing more fibre cores.

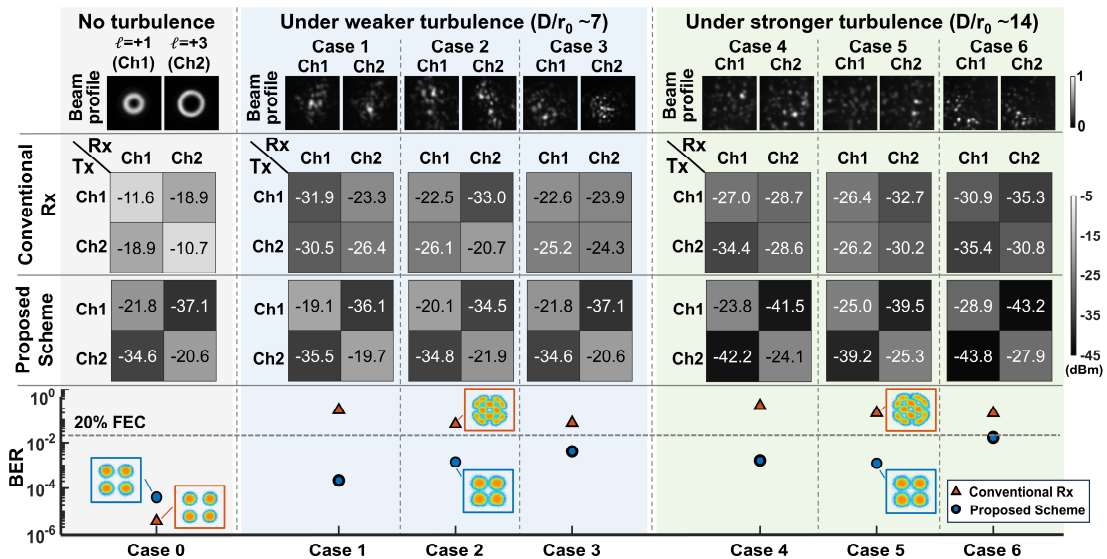


Fig. 4: Experimental performance under varied turbulence strengths. Top: Intensity profiles of the OAM modes under different turbulence levels. Middle: XT performance comparison between the conventional Rx and the proposed Rx. The values in the XT matrices are the output powers of the MCF. Bottom: BER performances and constellations of different Rx configurations.

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